INCH-POUND

MIL-DTL-22549E(SH)
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20 September 1968

DETAIL SPECIFICATION

VALVES, ANGLE, RELIEF, FOR GAS AND OXYGEN SERVICE (SIZES 2 INCHES NPS AND BELOW); NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers angle relief valves of pressure-tight spring housing (see 6.3.8) construction for gas and oxygen service (sizes 2 inches nominal pipe size [NPS] and below) for naval shipboard use.

1.2 Ratings. Valves covered by this specification have nominal pressure ratings of 400, 1500, 3000, and 6000 pounds per square inch gage (psig) (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-DTL-1183	-	Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS #C95800) Castings for Seawater Service

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-798	-	Design Evaluation Requirements for Valves for Naval Shipboard Use
MIL-STD-1330	-	Precision Cleaning and Testing of Shipboard Oxygen, Helium, Helium-Oxygen, Nitrogen, and Hydrogen Systems
MIL-STD-1622	-	Standard Practice for Cleaning of Shipboard Compressed Air Systems

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385884	-	Unions, Fittings and Adapters, Butt and Socket Welding, 6000 PSI, WOG, IPS
803-1385943	-	Unions, Silver Brazing, 3000 PSI, IPS, WOG, NPS, for UT Inspection
803-1385946	-	Unions, Bronze, Silver Brazing, WOD, for UT Inspection

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A351/A351M	-	Standard Specification for Castings, Austenitic, for Pressure Containing Parts
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B21/B21M	-	Standard Specification for Naval Brass Rod, Bar, and Shapes
ASTM B61	-	Standard Specification for Steam or Value Bronze Castings
ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B150/B150M	-	Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Valve description. The valves shall be self-contained, single seated, spring-loaded relief valves where the inlet pressure is sensed under and directly operates the spring loaded disc. Where operationally advantageous, other self-contained designs (such as inverted, expanded area, or pilot operated) that conform to the basic requirements of this specification may be permitted. However, all of these alternate designs shall be of the fail-open design and approval for their use is required by the Technical Authority (see 6.3.9).

3.3 Materials of construction. Materials shall be as specified herein. Selected materials shall withstand design pressures and shall be selected to prevent galling, seizing, or excessive wear between operating parts.

3.3.1 Gas service.

3.3.1.1 Body and spring housing. The body and spring housing shall be one of the following basic materials:

a. Corrosion resistant (stainless) steel (CRES) –Material grade CF8M of ASTM A351/A351M.

b. Aluminum bronze – Grade 9BHT of ASTM B148, alloy 2 of ASTM B150/B150M, or MIL-B-24480 (UNS C95800). When castings are used, the body shall be cast as a solid billet. Coring shall not be permitted.

c. Bronze – ASTM B61.

d. Naval brass – Alloy B of ASTM B21/B21M.

3.3.1.2 Springs. Springs shall be manufactured from CRES or carbon-steel. Where carbon-steel springs are used, they shall be electroplated with cadmium or zinc.

3.3.1.3 Non-metallic disc inserts, diaphragms, gaskets, and similar parts. All non-metallic parts shall be compatible with the fluid media specified (see 6.2). These parts shall also be compatible with both petroleum based and phosphate ester type fluids and shall be as approved by the Technical Authority.

3.3.1.4 Valve stems. Valves stems shall be manufactured from nickel-copper alloy (Monel, or similar) or CRES.

3.3.1.5 Nuts, bolts, washers, bushings, guides, bellows, and similar parts. All materials used in the construction of nuts, bolts, washers, bushings, guides, bellows, and similar parts shall withstand design pressures and marine environments and shall be in accordance with the material specification shown on the applicable approved drawing.

3.3.2 Oxygen service.

3.3.2.1 Body and spring housing. The body and spring housing shall be of nickel-copper alloy (Monel, or similar) in accordance with QQ-N-281 or ASTM A494/A494M.

3.3.2.2 Springs. Springs shall be manufactured from one of the following materials:

a. Bronze.

b. Nickel-copper alloy (Monel, or similar).

c. Copper-nickel alloy (70-30).

3.3.2.3 Valve stems. Valve stems shall be manufactured from nickel-copper alloy (Monel, or similar) or copper-nickel alloy (70-30).

3.3.2.4 Valve disc inserts. Valve disc inserts shall be manufactured from polytetrafluoroethylene (PTFE).

3.3.2.5 Diaphragms, O-rings, gaskets, and similar parts. Diaphragms, O-rings, gaskets, and similar parts shall be in accordance with the requirements specified in 3.3.1.3.

3.3.2.6 Nuts, bolts, bushings, guides, and similar parts. Nuts, bolts, bushings, guides, and similar parts shall be in accordance with the requirements specified in 3.3.1.5.

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3.3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction.

3.4.1 Pressure-temperature ratings. The pressure-temperature ratings for valves covered by this specification shall be as specified in 3.4.1.1 and 3.4.1.2.

3.4.1.1 Pressure-temperature rating (valve body [see 6.3.10]). The pressure-temperature rating for the valve body shall be in accordance with table I.

TABLE I. Pressure-temperature rating.

Nominal pressure rating (psi)	Design pressure (psig)	Design temperature (°F)	Proof pressure (psig)
400	480	165	600
1500	1800	165	2250
3000	3600	165	4500
6000	7200	165	9000

3.4.1.2 Pressure-temperature rating (spring housing). Unless otherwise specified (see 6.2), all parts of the pressure-tight spring housing shall be designed to withstand the higher of the following:

a. 150 percent of maximum system back pressure.
b. 100 psig.

The design temperature of these parts shall be as specified in table I.

3.4.2 Spring housings. There shall be one spring housing for each valve body of a particular nominal size and rating. It shall be designed to house all the springs required to span the applicable set pressure (see 6.3.5) ranges specified in 3.5.1. Valves without a pressure tight spring housing shall have a threaded vent port such that control of gas leakage through the vent port can be controlled in the same manner as control of the relief valve discharge.

3.4.3 End preparation.

3.4.3.1 Valves for gas service. All valves for other-than-oxygen service shall be provided with union end connections (threadpieces, tailpieces, and union nuts) in accordance with table II. Union threadpieces shall be cast or forged integral with the valve body or secured to the body with a straight thread and O-ring seal. Pipe threads shall not be used. Refer to documents listed in table II for pertinent dimensions of the end connections. The material for the union tailpieces shall be as specified (see 6.2). Materials selected for the union threadpiece and union nut shall be compatible with one another to prevent galling. Selected material for union threadpieces and union nuts is limited to those specified by the applicable drawing or specification listed in table II.

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TABLE II. End preparation (valves for gas service).

Nominal pressure rating (psi)	Drawing or specification
400	MIL-DTL-1183 1/
1500	803-1385946
3000	803-1385943
6000	803-1385884

NOTE:

1/ Where specified, union end connections in accordance with 803-1385946 may be used on 400-psig rated valves to give superior shock resistance.

3.4.3.2 Valves for oxygen service. Valves for oxygen service shall have 6-inch nipples welded directly to the valve body in both the inlet and outlet. The material used for the nipples shall be the same basic material as the body. End preparation shall be as specified (see 6.2).

3.4.4 Internal trim. A gasket or O-ring sealed seat ring shall be provided, which is either threaded-in or retained by a cage construction. The stem or stem and disc assembly shall be top guided. Guiding surfaces (bushings and posts) shall have the proper hardness, finish, concentricity, parallelism, clearances, length, and rigidity to prevent binding or seizing and to ensure proper seating under all operating conditions. These alignment requirements shall be maintained with interchangeable parts and under any tolerance stack-up condition. A non-metallic seating feature shall be incorporated in either the disc or seat.

3.4.5 Body construction. The valve shall be designed and constructed so that the seat will not become distorted relative to the disc and valve operation is not adversely affected by internal pressure.

3.4.6 Springs. Springs shall be designed so that they will not be compressed solid during any normal operation or adjustment of the valve. When removed and compressed solid, the spring shall not exhibit a permanent set exceeding 0.010 inch per inch of spring length, measured 10 minutes after release of the spring. Spring ends shall be squared and ground.

3.4.7 Threads. All threads shall be in accordance with FED-STD-H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used.

3.4.8 Bearing surfaces. Bearing surfaces of nut and bolt heads and their respective mating surfaces on the valves shall be finished machined.

3.4.9 Interchangeability. All parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance and shall not require selection, fitting, or machining of any kind.

3.4.10 Stem packing. There shall be no stuffing box on the valve stem.

3.4.11 Hand lifting device. Where specified (see 6.2), valves shall be designed so that they may be operated by hand for testing purposes with an inlet pressure of 75 percent of the set pressure. The necessary lever or tool shall be furnished as part of the valve.

3.4.12 Gagging device. Where required for system test purposes (see 6.2), a gagging device shall be specified to be supplied with the valve. Valves shall be capable of being gagged without alteration of the set point. The gagging screw shall be provided with a knurled or wing nut type head to discourage the use of wrenches when gagging the valve. The gagging device shall include a tag or other warning to this effect so as to minimize the possibility of overlooking its removal after test.

3.4.13 Accessibility. Valves shall be designed to permit adjustment and repair without removal from the line.

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3.4.14 Valve adjustment. Means shall be provided for adjusting the setting with the valve under pressure. The adjusting screw shall have right-hand threads so that clockwise rotation increases the set pressure. The adjusting device shall be provided with a locknut and cap or other means to prevent accidental change of adjustment.

3.5 Performance.

3.5.1 Range of adjustment. Valves shall be capable of being adjusted over the set pressure range (see 6.3.6) specified in table III. If required, more than one spring may be used to accomplish this.

TABLE III. Range of adjustment.

Nominal pressure rating (psig)	Minimum required set pressure (psig)	Maximum required set pressure (psig)
400	--	460
1500	460	1725
3000	1725	3450
6000	3450	6875

3.5.2 Operation. Relief valves shall operate over the entire flow range without chatter. Valves shall open with a clear sharp pop at the pressure for which they are set. Valve closure shall be clean and sharp when the inlet pressure is reduced to the blowdown pressure (see 6.3.4) after the valve has passed full rated flow or any intermediate flow.

3.5.3 Proof pressure. The valve shall show no signs of external leakage, permanent deformation, or structural failure when subjected to the proof test of 4.3.

3.5.4 Set pressure tolerance (see 6.3.7). The set pressure tolerance shall not exceed ±2 psi or ±2 percent of the set pressure, whichever is greater.

3.5.5 Accumulation (see 6.3.1). Valves shall be sized to pass the capacity specified (see 6.2) without permitting the inlet pressure to exceed the accumulation limit specified on figure 1. The valve shall show no signs of instability as a result of the accumulation test of 4.3.

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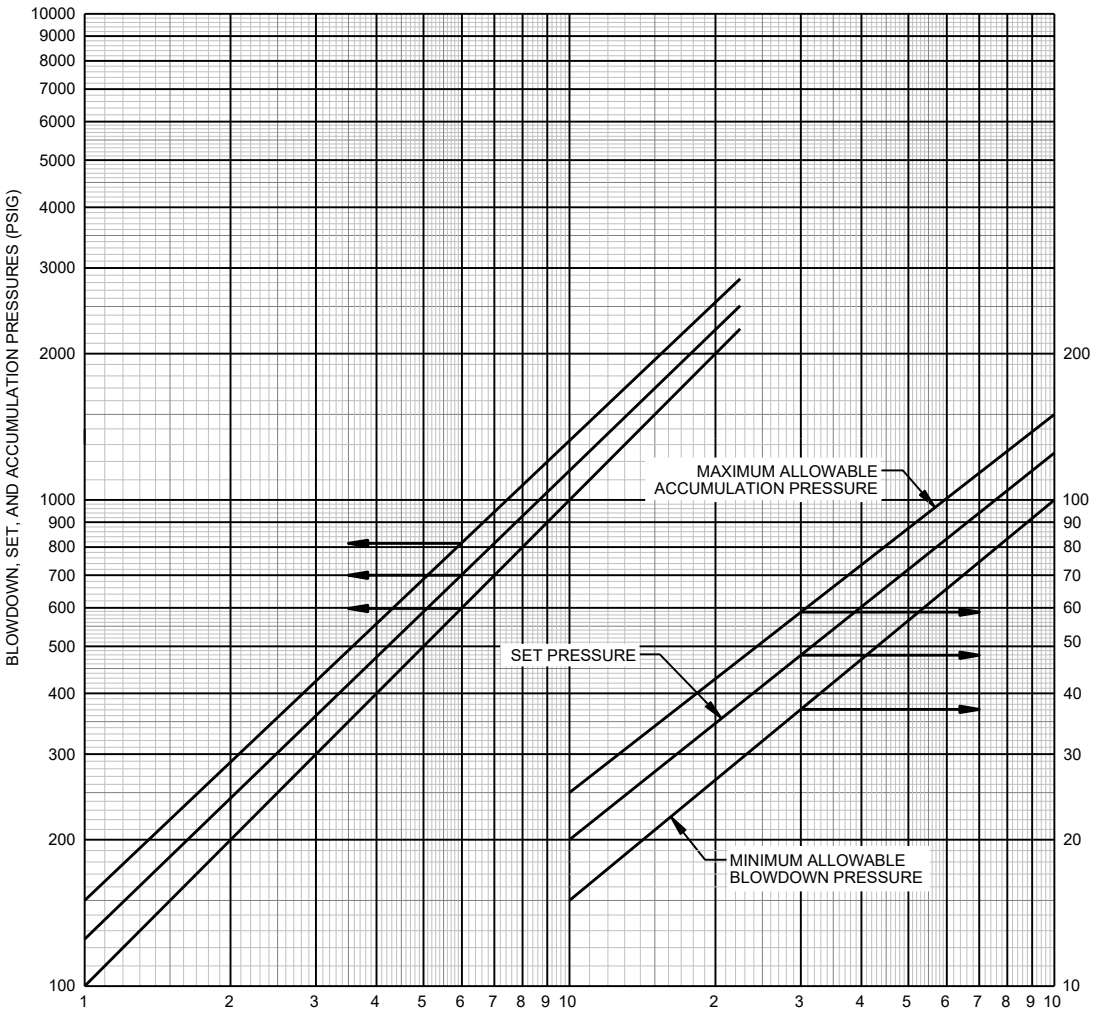


FIGURE 1. Relief valve characteristics.

3.5.6 Blowdown (see 6.3.3). Valves shall operate satisfactorily with a blowdown pressure not less than the minimum allowable blowdown pressure permitted by figure 1.

3.5.7 Seat tightness. With an inlet pressure at or above the minimum allowable blowdown pressure, the valve shall seat tightly. Allowable leakage under this condition shall not exceed 15 standard cubic inches per hour per inch of seat diameter.

3.5.8 Installation limitations. Unless otherwise specified (see 6.2), valve operation shall not be adversely affected by an inlet pressure loss of up to 25 percent of the relief valve blowdown, a discharge pressure buildup of up to 10 percent of the set pressure, or both.

3.5.9 Endurance. The valve shall show no signs of damage to the seating surface, instability, or leakage in excess of the amount specified in 3.5.7 and the set pressure shall remain within the tolerance specified in 3.5.4 after being subjected to the endurance test of 4.3.

3.6 Mechanical shock and vibration. Valves shall be designed to withstand testing in accordance with the mechanical shock requirements of MIL-DTL-901, the high impact (H.I.)-shock test guidance of MIL-STD-798, and the vibration requirements of MIL-STD-167-1.

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3.7 Marking.

3.7.1 Body markings. All valve bodies shall have the nominal pressure rating and manufacturer's name or trademark cast or forged integral with the valve body. When employed as a means of marking, metal stamping shall be permitted on valve body surfaces that are not subjected to high stress in service.

3.7.2 Identification plates. An identification plate made of CRES, brass, or anodized aluminum shall be permanently fastened to a part of the valve not subject to working pressure and shall contain the following information or a space therefor:

a. Manufacturer's name or trademark.

b. Body material composition.

c. Size of valve and rated capacity in standard cubic feet per minute (SCFM) at the applicable setting and accumulation.

d. Set pressure (psig), blowdown pressure (psig), and accumulation pressure (psig) (see 6.3.2).

e. Range of set pressure adjustment available with installed spring.

f. Installation limitations of valve (maximum permissible inlet pressure loss and maximum permissible back pressure buildup for which valve is designed).

g. Manufacturer's model or part number and drawing number.

h. Space for nine-digit allowance parts list number.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Degreasing.

4.2.1 Degreasing for compressed air (prior to assembly and testing). Prior to assembly for the tests specified in 4.3 and 4.4, valves for compressed air service shall be degreased in accordance with MIL-STD-1622.

4.2.2 Degreasing for oxygen and nitrogen service (prior to assembly and testing). Prior to assembly for the tests specified in 4.3 and 4.4, valves for oxygen and nitrogen service shall be degreased in accordance with MIL-STD-1330.

4.3 First article inspection. The first article inspection shall be performed on the first valve of the same size, design, pressure rating, setting, and capacity when a first article sample is required (see 3.1). This inspection shall include the examination and tests specified in table IV. Acceptance criteria shall be as specified in table IV.

TABLE IV. Verification tests and examinations.

Test	Test conditions	Purpose of test	Acceptance criteria
Visual examination	--	To verify conformance with the requirements of this specification.	Complete conformance.
Material examination	Valve shall be examined to determine conformance with applicable material specification. For non-pressure-containing parts, certification by manufacturer may be accepted in lieu of examination.	To verify conformance of material to drawing and specification requirements.	Complete conformance.

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TABLE IV. Verification tests and examinations – Continued.

Test	Test conditions	Purpose of test	Acceptance criteria
Proof test	Valve gagged shut or disc and spring assembly removed and seat blanked off. Water or air at 1.5 times pressure rating applied to inlet.	To test strength and soundness of pressure-containing envelope.	No external leakage, permanent deformation, or structural failure.
Set pressure, blowdown, and seat tightness test [1/]	Air inlet pressure increased until valve lifts. Inlet pressure reduced until valve reseats. Check for leakage over a 3-minute period with an inlet pressure equal to the minimum blowdown pressure permitted by figure 1.	To determine set point and blowdown setting of valve. To test for seat tightness at the minimum allowable blowdown pressure.	Blowdown (see 3.5.8). Seat tightness (see 3.5.7). No damage to seating surfaces. No instability.
Accumulation test [2/]	Air inlet pressure increased until valve passes rated flow.	To verify relief valve capacity. To check for instability over operational range of valve.	Valve to pass rated flow without exceeding accumulation permitted by figure 1. No instability.
Set pressure repeatability and endurance test [3/]	Using air, cycle valve 50 times. After every 10 cycles, check for leakage.	To verify that set pressure repeatability is within allowable limits. To verify ability of valve to withstand repetitive cycling.	Set pressure repeatability (see 3.5.4). Seat tightness (see 3.5.7). No damage to seating surface caused by cycling impact. [4/] No instability.

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TABLE IV. Verification tests and examinations – Continued.

Test	Test conditions	Purpose of test	Acceptance criteria
H.I. shock test 5/	Grade A, class I of MIL-DTL-901 and H.I. shock test guidance of MIL-STD-798. Valve shall be pressurized during test.	To determine ability of the valve to withstand H.I. mechanical shock.	No structural damage or degradation to performance capability.
Vibration test 5/	Type I of MIL-STD-167-1.	To determine effect of environmental vibration.	No resonant frequency 0 to 33 hertz. No structural damage or degradation to performance.

NOTES:

1/ The test setup shall impose an inlet pressure loss and outlet pressure buildup equal to the maximums specified in 3.5.8 (25 percent of the blowdown and 10 percent of set pressure). Where greater losses are specified (see 6.2), they shall be imposed instead.

2/ Accumulation test may be waived provided that capacity data, obtained and certified in accordance with section VIII of the ASME Boiler and Pressure Vessel Code, is available.

3/ This test may be waived provided that certified test data is available showing that the same basic design and size has previously passed these tests.

4/ Any damage caused by foreign particle entrapment on the seating surfaces shall be repaired prior to acceptance of the valve.

5/ Shock and vibration tests shall be required only when specified (see 6.2). Tests will not be required if the valve has been previously approved for shock and vibration.

4.4 Conformance inspection. Conformance inspection shall include the visual and material examinations and the proof, set pressure, blowdown, and seat tightness tests of table IV.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. All valves covered by this specification are intended for use in shipboard gas and oxygen systems. Current stock of relief valves on Navy ships may be used until depleted; future requisitions for relief valves should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Pressure rating required (see 1.2).

c. When first article inspection is required (see 3.1).

d. Line media (see 3.3.1.3).

e. Spring housing design pressure, if other than specified herein (see 3.4.1.2).

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f. Quantity and pipe size required (see 3.4.2).

g. Material required for the union tailpiece (see 3.4.3.1).

h. Nipple welding end preparation for oxygen valves (see 3.4.3.2).

i. If a hand lifting device is required to be supplied with valve (see 3.4.11).

j. If a gagging device is required to be supplied with valve (see 3.4.12).

k. Set pressure and capacity required (see 3.5.5).

l. If required by intended installation, allowable inlet pressure loss greater than 25 percent of relief valve blowdown (see 3.5.8 and table IV).

m. If required by intended installation, allowable discharge pressure buildup of greater than 10 percent of set pressure (see 3.5.8 and table IV).

n. Packaging requirements (see 5.1).

6.3 Definitions. The following definitions are applicable to this specification.

6.3.1 Accumulation. The increase in pressure above the set pressure required to pass rated flow. Expressed in psi or as a percent of the set pressure.

6.3.2 Accumulation pressure. The set pressure plus the accumulation. Expressed in psig.

6.3.3 Blowdown. The decrease in pressure below the set pressure required for the valve to reseat. Expressed in psi or as a percent of the set pressure. The accumulation and blowdown establish the operating band of the relief valve at a particular setting.

6.3.4 Blowdown pressure. The set pressure minus the blowdown. Expressed in psig.

6.3.5 Set pressure. The pressure at which the valve pops. Expressed in psig.

6.3.6 Set pressure range. The range over which the set pressure can be adjusted with the installed spring.

6.3.7 Set pressure tolerance. The permissible plus or minus deviation from the specified set pressure. Expressed in psi or as a percent of the set pressure.

6.3.8 Spring housing. All portions of the pressure-containing envelope subject to outlet or discharge pressure.

6.3.9 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.3.10 Valve body. All portions of the pressure-containing envelope subject to inlet pressure.

6.4 Subject term (key word) listing.

Compressed air
Nitrogen service
Pressure release

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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